

§1 $\frac{d}{dx} \int_0^x (x-t) f(t) dt = \frac{d}{dx} (x \int_0^x f(t) dt - \int_0^x t f(t) dt) = \int_0^x f(t) dt + x f(x) - x f(x) = \int_0^x f(t) dt$

Hessian $\Delta = Ac - \beta^2 > 0$ $A > 0$ 极小.

$A > 0$ 极大

$\Delta < 0$ 鞍点.

$\Delta = 0$ 不确定

n 主量子数 / rule

$0 \leq l < n$

$-l \leq m \leq l$

$\sum_{l=0}^{n-1} (2l+1) = n^2$

$\langle n | x | n \rangle \propto \sqrt{n} \delta_{n,n-1}$

$\sqrt{n+1} \delta_{n,n+1}$

§2 M1 $\lim_{x \rightarrow 0} \frac{\cos(\sin x) - 1 + \frac{1}{2}x^2}{x^4}$

(1) $\frac{1}{2} u = \sin x$

$\lim_{x \rightarrow 0} \frac{1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 - 1 + \frac{1}{2}x^2}{x^4}$

$\lim_{x \rightarrow 0} \frac{1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 + o(x^4) - 1 + \frac{1}{2}x^2}{x^4} = \frac{5}{24}$

若展开到 x , $\frac{1}{2}u^2$ 和 $\frac{1}{24}u^4$ 项, $o(x^4)$ 就剩有 x^4 项, 而 $(x - \frac{1}{6}x^3)$ 项中也有 x^4 出现, 至外留到最高项

(2) $V(x) = V_0 (\frac{a^2}{x^2} - \frac{2a}{x})$

$V'(x) = -\frac{2V_0 a^2}{x^3} + \frac{2aV_0}{x^2}$

$U(x) = 0$ 平衡位置.

$\frac{2V_0 a^2}{2x^3} = \frac{2aV_0}{x^2}$

$x = a$ $x_0 = a$

$V''(x) = \frac{6V_0 a^2}{x^4} - \frac{4aV_0}{x^3}$

$k_{eff} = V''(x_0) = \frac{6V_0}{a^2} - \frac{4V_0}{a} = \frac{2V_0}{a^2}$

$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{2V_0}{ma^2}}$

M2 (1) $D_n = 2\cos\theta D_{n-1} - D_{n-2}$ (3)

$D_n - 2\cos\theta D_{n-1} + D_{n-2} = 0$

$\lambda^2 - 2\cos\theta \lambda + 1 = 0$

(2) $D_1 = 2\cos\theta$

$D_0 = 1$

$D_2 = 4\cos^2\theta - 1$

$\lambda_1 = \cos\theta + i\sin\theta = e^{i\theta}$
 $\lambda_2 = \cos\theta - i\sin\theta = e^{-i\theta}$

通解 $D_n = C_1 \lambda_1^n + C_2 \lambda_2^n = C_1 e^{in\theta} + C_2 e^{-in\theta}$

$D_0 = C_1 + C_2 = 1$

$D_1 = C_1 \lambda_1 + C_2 \lambda_2 = 2\cos\theta$

$D_2 = C_1 \lambda_1^2 + C_2 \lambda_2^2 = 4\cos^2\theta - 1$

$C_1 \lambda_1 + C_2 \lambda_2 = 2\cos\theta$

$\Rightarrow \begin{cases} C_1 = \frac{\lambda_2}{\lambda_1 - \lambda_2} \\ C_2 = \frac{\lambda_1}{\lambda_1 - \lambda_2} \end{cases}$

$C_1 \cos\theta - C_1 i\sin\theta + C_2 \cos\theta + C_2 i\sin\theta = 2\cos\theta$
 $(C_1 + C_2)\cos\theta + (C_2 - C_1)i\sin\theta = 2\cos\theta$
 $\Rightarrow \begin{cases} C_1 + C_2 = 2 \\ C_2 - C_1 = 0 \end{cases}$

$C_1 = \frac{\lambda_1^n - \lambda_2^n}{\lambda_1 - \lambda_2} = \frac{e^{in\theta} - e^{-in\theta}}{e^{i\theta} - e^{-i\theta}} = \frac{e^{i(n+1)\theta} - e^{-i(n+1)\theta}}{e^{i\theta} - e^{-i\theta}} = \frac{\sin((n+1)\theta)}{\sin\theta}$

Q1

(I) $n \cdot 0 \leq l \leq n-1 \quad -l \leq m \leq l$

(II), $g_n = \sum_{l=0}^{n-1} (2l+1) = \frac{1 + (2n-1+1)}{2} \cdot n = n^2$

计入自旋时 同一轨道一个上旋 一个下旋 2种状态 $2n^2$

(III)

π 对称性本征

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Q2

(I), $\hat{a} = \sqrt{\frac{m\omega}{2\hbar}} (\hat{x} + \frac{i}{m\omega} \hat{p}) \quad \hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}} (\hat{x} - \frac{i}{m\omega} \hat{p})$

$[\hat{a}, \hat{a}^\dagger] = \frac{m\omega}{2\hbar} \left[\left(\hat{x}^2 + \left(\frac{i}{m\omega} \right)^2 \hat{p}^2 - \hat{x} \frac{i\hat{p}}{m\omega} + \frac{i}{m\omega} \hat{p} \hat{x} \right) - \left(\hat{x}^2 + \left(\frac{i}{m\omega} \right)^2 \hat{p}^2 - \frac{i\hat{p}\hat{x}}{m\omega} + \frac{i\hat{x}\hat{p}}{m\omega} \right) \right]$
 $= \frac{m\omega}{2\hbar} \left(\frac{2i}{m\omega} \hat{p}\hat{x} - 2i\frac{\hat{x}\hat{p}}{m\omega} \right) = -\frac{i}{\hbar} [\hat{x}, \hat{p}] = 1$

(II), $\hat{a}|n\rangle = \sqrt{n}|n-1\rangle$

$\hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$

$\hat{a}|0\rangle = 0$

$\langle n' | \hat{x} | n \rangle = \langle n' | \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} + \hat{a}^\dagger) | n \rangle$

$= \sqrt{\frac{\hbar}{2m\omega}} \langle n' | \sqrt{n}|n-1\rangle + \sqrt{\frac{\hbar}{2m\omega}} \langle n' | \sqrt{n+1}|n+1\rangle$

$= \sqrt{\frac{\hbar}{2m\omega}} (\sqrt{n} \delta_{n',n-1} + \sqrt{n+1} \delta_{n',n+1})$

13)

$\delta n, n-1 \Rightarrow n' = n-1$

$\delta n, n+1 \Rightarrow n' = n+1$

$\Rightarrow \Delta n = \pm 1$ 时不为零. 否则选择归一化.

$\langle n | \hat{x} | n \rangle = \sqrt{\frac{\hbar}{2m\omega}} (\sqrt{n} \delta_{n,n-1} + \sqrt{n+1} \delta_{n,n+1}) = 0$

字

在

$\int_{-\infty}^{\infty} x |\psi|^2 dx$

$\langle n | \hat{x} | n \rangle = \int x |\psi|^2 dx$

是奇函数 偶函数

不为零. $x |\psi|^2$ 是奇函数. 积分为0.